Ferroelectric Nematic Glass-Based Silicon Photonics Modulator for Net 400 Gbps IM/DD Transmission

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Abstract: We demonstrate a record net 400Gbps IMDD data transmission using a hybrid-integrated silicon and ferroelectric nematic glass modulator. Transmission of 200Gbaud PAM6 and 168Gbaud PAM8 is achieved below SD25-FEC over 100m of SSMF. ©2026 The Author(s)

1. Introduction

The relentless growth of data traffic in communication networks continues to drive the demand for optical modulators with higher speeds, lower power consumption, and smaller footprints. Silicon photonics has emerged as a leading platform for realizing high-performance photonic integrated circuits, offering the potential for scalable and cost-effective solutions. However, achieving PN junction-based modulators with bandwidths exceeding 100 GHz remains extremely challenging. To enhance modulator performance, various electro-optic materials are being actively explored for silicon integration, including ferroelectric crystals (e.g., TFLN, BTO), III-V semiconductors, and electro-optic polymers [1-4].

Polaris Electro-Optics' proprietary ferroelectric nematic glass (FenGlass™) is a new class of ferroelectric material that offers a compelling alternative, leveraging the large Pockels effect of these materials combined with easy integration onto silicon, compatibility with efficient device structures, and the ability to continuously engineer larger electro-optic coefficients [5]. Early prototypes of Si-FenGlass hybrid modulators demonstrated the potential of these structures with data rates up to 100 Gbps [6]. In this work, using a Si-FenGlass hybrid modulator with a 0.5 mm phase shifter, we report 176 Gbaud PAM4 below the 6.7% overhead hard-decision forward error correction (HD-FEC) threshold, enabling a net rate of 330 Gbps over 100 m of standard single-mode fiber (SSMF). Further, we demonstrate a net rate exceeding 400 Gbps with 200 Gbaud PAM6 and 168 Gbaud PAM8 signaling, below the 25% overhead soft-decision forward error correction (SD-FEC) threshold over 100 m of SSMF. These results show that the performance of Si-FenGlass hybrid modulators is extended to 200 Gbaud, highlighting the purely electronic origin of the Pockels effect and the promise of this platform for next generation electro-optic modulators.

2. Si-FenGlass Hybrid Modulator Design and Characterization

The modulator evaluated in this work integrates FenGlass into doped silicon slot waveguides within a Mach–Zehnder interferometer (MZI) architecture [5–7]. Both phase-shifting arms employ slot waveguides operating in the TE0 mode at the C-band, as shown in Fig. 1(d). To support high-speed performance, the single modulator employs a GSGSG traveling-wave electrode configured for push–pull drive. The cross-section of the modulator is shown in Fig. 1(c). The phase-shifter length is 0.5 mm, and a 50 Ω on-chip termination is used. The half-wave voltage V_π of this device was measured using a 1 MHz sinusoidal driving signal and is 6.7 V. Fabrication was carried out on a 220 nm silicon photonics platform using a standard commercial foundry process. The FenGlass material was dispensed as an ink and processed to form the ferroelectric glass phase. A ferroelectric monodomain was then created along the phase shifter by reorienting the polar axis with a small external DC field to activate the device [5]. Fig. 1(b) shows the received RF spectrum of the device operating at 224 Gbaud. The spectrum shows the combined frequency response of the FenGlass modulator, the Arbitrary Waveform Generator (AWG), the 67 GHz RF probe, and the photodetector.

The linearity of the driving voltage to the Electro-Optic (EO) phase transfer function is a prerequisite for advanced modulation formats (e.g., coherent) and Linear Pluggable Optics (LPO) [8,9]. We tested the linearity by measuring the EO phase generated by the FenGlass modulator as a function of increasing amplitude of the applied sinusoidal driving signal. The phase is extracted by fitting the output waveform to the standard MZI transfer function model. As the phase approaches and exceeds π, the fit becomes insensitive to the optical modulation amplitude (OMA) and allows an accurate extraction of the phase. Fig. 2(a) reports the extracted EO phase versus driving amplitude. Very linear behaviour is observed across a wide range of voltages, up to the maximum of 20 Vpp (limited by the instrument), corresponding to an electric field of ~0.5 MV/cm in the material. No breakdown was observed. A $V_\pi L$ of 3.4 V·mm

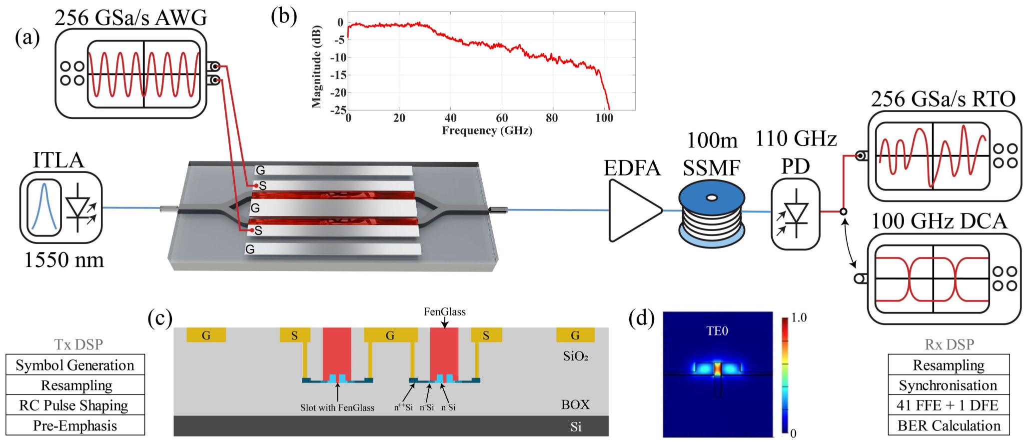

Fig. 1: (a) Schematic of the experimental setup and employed DSP; (b) Received RF spectrum of a 224 Gbuad PAM4 signal (ROP: 7 dBm); (c) Cross-sectional schematic of the FenGlass-MZM. (d) Optical mode field distribution of a Si-FenGlass slot waveguide.

was obtained from the phase extraction technique, corresponding to an estimated in-device EO coefficient r_{33} of 27 pm/V.

Flatness of the EO response at low frequency is important to avoid baseline wandering in the eye, which can impair the receiver's ability to threshold data. Some of the EO materials considered today for high-speed modulators suffer from significant low-frequency variation in EO efficiency due to low-speed contributions to the EO effect (piezoelectric, ionic, etc.) [10]. We characterized the FenGlass modulator with a 1 mm long phase shifter over the frequency range from 50 mHz to 1 MHz using a DC-coupled sinusoidal driving signal and extracted the phase from the generated waveform at the output of the MZI, as presented in Fig. 2(b). The V_π of the device remains substantially flat, indicating a lack of spurious contribution to the Pockels electro-optic effect and the V_π of the device changed by approximately 10% across more than 7 decades.

High reliability is a fundamental requirement in data centers. Thus, the devices were tested for more than 1000 hours at a temperature of 70 °C with an applied driving voltage of 4 Vpp and input optical power of approximately 10 dBm to the modulator. The FenGlass material integrated on the silicon photonic chip was exposed to the environment without encapsulation of any kind, indicating a promising degree of intrinsic photochemical and thermal stability. The trend of the EO phase vs stress measurement is well behaved and fits a power-law model, projecting a decrease of less than 20% over a period of 10 years, as shown in Fig. 2(c). The observed decrease in EO phase was fully recovered after re-aligning the material in the slot, indicating no permanent degradation effect.

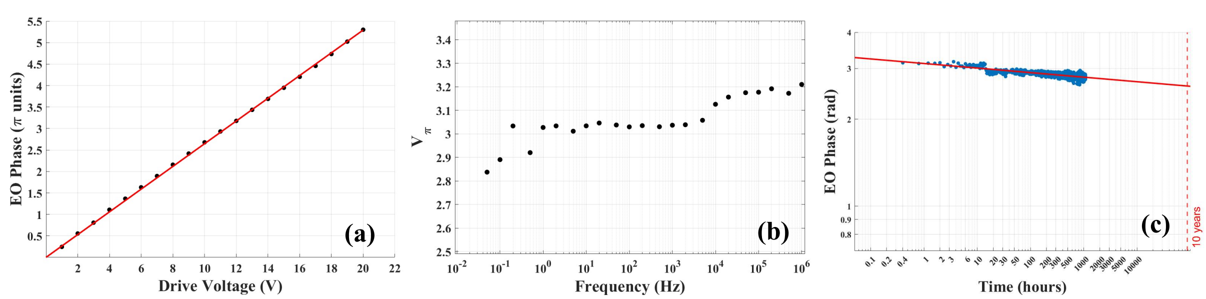

Fig. 2: (a) Linearity of EO phase with applied voltage (@ 1MHz). (b) V_π vs frequency in the low frequency ranges from 50mHz to 1MHz. (c) EO phase recorded during High-temperature operating life (HTOL) stress test at 70°C and >1000hrs for a non-encapsulated modulator.

3. Transmission Experiment

Fig. 1(a) shows the experimental setup, including the DSP stack used for data transmission. PAM4, PAM6 and PAM8 signals with a pattern length of 2^{19} are generated by Mersenne Twister and resampled to match the Keysight M8199B AWG operating at a sample rate of 256 GSa/s. A raised cosine filter is applied to the symbols for pulse shaping, and the pre-emphasis filter is employed at the transmitter to compensate for the loss from the RF connections and the frequency response of the AWG. The modulated symbols are then loaded onto the differential channel of the AWG with a voltage swing of 2.4 Vpp. The differential RF signals are applied to the modulator using a 67 GHz RF probe with the GSGSG configuration.

An Integrable Tunable Laser Assembly (ITLA) with output power of 15 dBm at 1550 nm is used as the optical source. The laser wavelength is tuned such that the modulator operates at the quadrature point to ensure linear modulation. The modulated signal is then amplified using an Erbium-Doped Fiber Amplifier (EDFA) to compensate for the loss incurred in the edge coupling, the strip waveguide to slot waveguide converter, and through the phase shifter. The optical signal is then transmitted over a 100 m SSMF and is received by a photodiode (PD) with a bandwidth of 110 GHz, which converts the received optical signal to an electrical signal. The electrical signal from the PD is then sampled by a 256 GSa/s real-time oscilloscope (RTO) for analog-to-digital conversion. The gain of the EDFA is adjusted to ensure a received optical power (ROP) of 7 dBm at the PD, providing sufficient signal swing for detection by the RTO. No transimpedance amplifier (TIA) is used between the PD and the RTO.

The digitized waveforms are downloaded from the RTO and processed offline. The signal is resampled to 2 samples per symbol and synchronized with the transmitted signal. A Feed-Forward Equalizer (FFE) with 41-taps and a Decision Feedback Equalizer (DEF) with 1-tap are applied to the signal. The equalized signal is then mapped back to a binary sequence for comparison with the transmitted sequence to calculate the Bit-Error Rate (BER).

For eye diagram analysis, the PD is connected to the Keysight 110GHz Digital Communication Analyzer (DCA) to capture the waveform. A Short Stress Pattern Random-Quaternary (SSPRQ) signal with a pattern length of 2^{16} is generated using the AWG and transmitted through the Si-FenGlass hybrid modulator.

Fig. 3(a) shows the measured BER for PAM4 (blue), PAM6 (red), and PAM8 (yellow) over 100 m of SSMF. Under the 6.7% overhead HD-FEC threshold, we achieve net rates of 330 Gbps at 176 Gbaud PAM4 and 136 Gbaud PAM6. For PAM8 signalling, meeting the HD-FEC threshold at high baud rates is difficult, indicating a higher SNR requirement for PAM8 [11]. At the 25% overhead SD-FEC threshold, a net rate of 358 Gbps is achieved at 224 Gbaud PAM4. We could not increase the baud rate further due to the limited frequency response of the AWG and the high RF loss introduced by the 67 GHz RF probe. The highest net rate achieved is 414 Gbps at 200 Gbaud PAM6. A net rate of 403 Gbps is also achieved with 168 Gbaud PAM8.

Fig. 3(b) shows an eye diagram from the Si-FenGlass hybrid modulator at 176 Gbaud PAM-4 using a SSPRQ sequence. The eye is recovered using 41-taps of FFE and 1-tap of DFE.

4. Conclusion

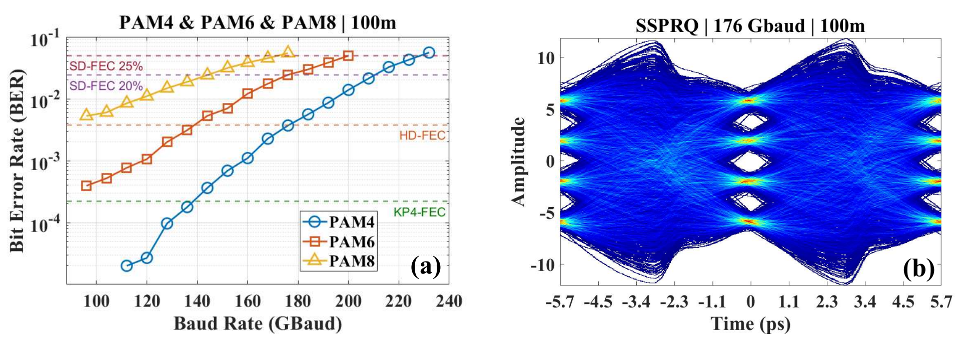

Fig. 3: (a) BER vs. Baud rate for PAM4 (blue), PAM6 (red), and PAM8 (yellow) at 100m SSMF using 41-taps FFE + 1-tap DFE. (b) Eye diagram of 176 Gbaud SSPRQ signal equalized with 41-taps + 1-tap DFE.

In this work, we demonstrated the first high-speed data transmission using a hybrid silicon–ferroelectric nematic glass (Si-FenGlass) modulator. By leveraging the strong Pockels effect of FenGlass integrated into silicon slot waveguides, we achieved high performance below the SD-FEC thresholds at net rates exceeding 400 Gbps. We achieve a net rate of 330 Gbps with 176 Gbaud PAM4 under the HD-FEC threshold. Net rates of 414 Gbps and 403 Gbps are achieved with 200 Gbaud PAM6 and 168 Gbaud PAM8 signaling under the 25% OH SD-FEC threshold, respectively. The device shows linear EO phase response up to large drive voltages and a flat low-frequency response, indicating a purely electronic Pockels mechanism. These results highlight the robustness and scalability of FenGlass as an electro-optic material for silicon photonics and enabling the next generation of high-speed data transmission.

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